Specific Plant Data For 03xAcademic Block At CUI Attock Campus (400 kW)		
EQUIPMENT	**SPECIFICATIONS**	**Yes/No**
Inverters & Hybrid Formation	Three Phase Grid Tied	
	Hybrid Inverter	
	Hybrids formation priority: 1.Same Row,2.Adjacent Row	
	Hybrid Voc: Must be within MPPT range at -10 to + 80 Degree Celsius cell temperature	
	Anti-Potential Induced Degradation Module (PID)	
	Sheds and enclosures for outdoor installation	
	50 to 60 Hz Frequency Output	
	CEC Efficiency of 95.5% or above	
	IP66 rated	
	12 Year Plus Warranty	
	THD < 3%	
	IEC 62109/UL1741 Compliant Safety Design	
	Transformer-less modular Design	
	Ambient Operating Temperature Range: -25 ºC to 60 ºC	
	Multiple MPPT Trackers	
	DC/AC ratio not more than 1.1	
	Residual Current Monitoring	
	DC Insulation Resistance Detection	
	Generator Synchronization module	
	AC and DC Surge Arrestors: Type II or better or certified per IEC 61643-11 or 20 KA for AC and DC or equivalents	
	AC Overcurrent Protection	
	DC reverse polarity protection	
	PV Array Hybrid Fault Monitoring	
	Anti-Islanding Protection	
	WiFi and Cloud Communication	
	USB, Modbus, RS485	
	S.R.O 604 Compliant	
AC,DC Cables, Earthing wires &Cable Trays	Copper Cables with purity of 99.99%	
	UL 4703 compliant Low Smoke Halogen Free	
	DC cable size : 06 mm2 from array of panel to DB DC cable size : 16 mm2 from array DB to Main DB AC cable size : As per attached Drawing	
	XLPO Cables for operation under elevated temperatures rated for 1500 V	
	Conduits shall have 30% or higher spare capacity	
	Preparation of Site and all Civil Works	
	Seamless Terminations	
	Perforated Cable Trays of 16 Gauge or thicker	
	Avoid Direct Sunlight and Direct Clamping of Cables	

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Specific Plant Data For 03xAcademic Block At CUI Attock Campus (400 kW)		
EQUIPMENT	**SPECIFICATIONS**	**Yes/No**
	Cables should not be physically accessible for routine works	
	Labeling of all Hybidsequipment shall be mandatory	
	06 mm^2 earth wire for equipotential bonding	
	Cables to be laid in UPVC/corrugated PVC pipes and cable trays	
System Design and Installation/As Built Drawings	Energy Yield Assessment	
	Module Layout Drawings as per the drawings attached and Referred	
	Cable Layout Plans	
	As built Drawings	
Breakers, Surge Arrester & LV Panel	Circuit Breakers	
	Changeover Switches	
	Class-II Surge protection separate for AC & DC sides	
	Separate DC and AC Earthing each with 5 Ohms or lesser resistance	
	Non-Isolated Lightning Arrestors	
	Protection against lightning currents of 31 kA or higher	
	In case of LV panel installation, the panel shall be IP65 rated with galvanized material of at least 16 gauge with protection Scheme as per IEC Standard 62548, having Input and Output Circuit Breakers as per IEC Standards, Bus Bar Current Bearing Capacity as per requirement, Isolators for Bus Bars and AC Grounding Bus Bar.	
	The panel shall have Digital Ammeter and Voltmeter, RYB indicator with Breather Vents on both Sides of cabinets with RAL Color: 6021	
Monitoring and Control System	Centralized Monitoring and Logging of array level input, System Output	
	Communication through the following: 1. RS485/RS232/TTL/USB for onsite data collection/monitoring 2. GPRS/4G for real-time data transfer to Client's defined server (subscription of GPRS/4G to be paid by the Contractor) 3. Wifi/Bluetooth Module for localized wireless communication 4. Alarm Messages (SMS & Email) for Fault/ Grid Outage / Generator Outage / Inverter Outage etc. 5. Daily Generation data 6. Each Inverter data 7. Monthly production data 8. Yearly production data 9. Total production data	
Net Metering Application and	Compliance with all prevailing regulations set forth by NEPRA/IESCO/Punjab Energy Department for net metering and safety purposes	
	Load flow studies	

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Installation of Bidirectional Meter	Preparation and submission of application for net metering	

Specific Plant Data For 03xAcademic Block At CUI Attock Campus (400 kW)		
EQUIPMENT	**SPECIFICATIONS**	**Yes/No**
Inverter Data Logger Collection of Data	Operating status	
	Vdc	
	Vac	
	Pdc	
	Pac	
	Grid Status	
	Generator Status (from Genset Sync. Module)	
	Fault Detection	
	Daily Energy Supplied by PV System	
	Total Energy Supplied by PV System	
	Plant Performance Data Check: Every 15 Seconds or lesser	
	Total Energy Supplied by Genset (from Genset Sync. Module)	
Synchronization Modules for GENSETS AND INVERTERS	GENSETS syncing modules Compatible with Diesel generators and Proposed Inverters	

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Appendix-K

BILL OF QUANTITIES

A. Preamble

1. The Bill of Quantities shall be read in conjunction with the Conditions of Contract, Specifications and Employer Design given in the Bidding Documents.

2. The rates and prices entered in the priced Bill of Quantities shall, except in so far as it is otherwise provided under the Contract include all costs of Contractor's plant, labour, supervision, materials, execution, insurance, profit, taxes and duties, together with all general risks, liabilities and obligations set out or implied in the Contract. Furthermore, all duties, taxes and other levies payable by the Contractor under the Contract, or for any other cause, as on the date of deadline for submission of Bids, shall be included in the rates and prices and the total Bid Price submitted by the Bidder.

3. A rate or price shall be entered against each item in the priced Bill of Quantities, whether quantities are stated or not. The cost of items against which the Contractor will have failed to enter a rate or price shall be deemed to be covered by other rates and prices entered in the Bill of Quantities.

4. The whole cost of complying with the provisions of the Contract shall be included in the items provided in the priced Bill of Quantities, and where no items are provided, the cost shall be deemed to be distributed among the rates and prices entered for the related items of the Works.

5. General directions and description of work and materials are not necessarily repeated nor summarised in the Bill of Quantities. References to the relevant sections of the Bidding Documents shall be made before entering prices against each item in the priced Bill of Quantities.

6. Provisional sums included and so designated in the Bill of Quantities shall be expended in whole or in part at the direction and discretion of the Employer.

7. Any increase in quantities (other than PV module capacity of 400 kW + 2%) during design review stage shall not impact the quoted price(s).

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BILL OF QUANTITIES

The Bill of Quantities is given below for Bidders to price.

	BOQ FOR SOLARIZATION OF 03xAcademic Block At CUI Attock Campus (400 kW)					
No.	**ITEM(s)**	**Description**	**Unit**	**Minimum Quantity**	**Rate (Rs.)**	**Item / Service Price / Cost Inclusive of all Applicable Taxes (PKR)**
	A. SUPPLY OF GOODS FOR 03xAcademic Block At CUI Attock Campus (400 kW) SOLAR PV SYTEM					
	A. CIVIL WORKS					
1	PV Mount Structure for a) Computer Science Block(150kw) b) Math's/Management Science Block(150kw) c) Electrical & Computer Engineer Block(100kw)	Supply and fixing PV Module Site preparation works including clearance and dismantling of excavation/filling/compaction. Wind speed = 160km/h, azimuth as per site. (UV resistant,120 micron of zinc coating), 12 SWG i.c 2 coat of enamel paint, and one coat of red oxide of approved color or as per drawing given in Employer's Design. complete in all respect. Or as directed by engineer Incharge.	Sq M	2250		
2	Cable Tray	Supply and fixing of 16 SWG Galvanized iron with flat cover, perforated downside, complete fitting accessories i.e nut bolts, supporting angle iron etc, size (W*D) inch 8*4 for laying string wires or as per International Standard or complete in all respect.	M	200		
3	UPVC pipe	Supply and fixing of 75mm dia pipe B class for laying string wires PV area to inverters including sockets,	M	950		

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		i/c excavation, backfilling complete in all respect. Or as directed by engineer incharge.				
4	UPVC pipe	50mm dia pipe B class for laying string wires PV area to inverters including socketsi/c excavation, backfilling complete in all 5respect. Or as directed by engineer incharge.	M	700		
5	UPVC pipe	25mm dia pipe B class for laying string wires PV area to inverters including socketsi/c excavation, backfilling complete in all respect. Or as directed by engineer Incharge.	M	500		
6	PCC(1:3:6)	Clearing, grabbing, Dressing, leveling, compacting and Providing and laying 1:3:6 (1 cement 3 Lawrencepur sand and 6 coarse aggregate crushed graded boulders (machine broken) are used) cement concrete using graded gravel (bajri) 1inch (75 mm) and down gauge in foundation including levelling, compacting and curing etc. complete in all respect. Or as directed by engineer Incharge.	Cubic M	210		
7	Tuff Paver	Providing and laying in floor C.C. 1:3:6: tuff paver 8"x4"x2-1/2" thick of approved design and colour pattren (average strength 7000 psi) laid on sand cushion filling of joint with sand and warring etc. complete as per direction of Engineer Incharge. (the cost of sand cushion is included)	Sq M	2250		
Subtotal A (inclusive of all applicable taxes including GST of Services at Sr. No. 1 to 7) = PKR						

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	B. Electrical Works					
1	PV module	Tier 1 Manufacturer according to Bloomberg latest issued rankings with IEC 61215 Compliant, not less than 650 W N-Type and Half Cut PV modules or better	KW	400		
2	Hybrid Inverter 150kw 150kw 100kw	3 phase On grid Inverter with Remote Monitoring Devices	KW	400		
3	Communication Box	Data Logger complete in all respect as per international standard or NPERA/IESCO.	No.	As per inverter Configuration		
4	Genset Synchronization Module	Genset Auto Synchronization Hybrid Gen-set Controller/Solar-Diesel Integration Controller compatible with Inverter and Generator, factory fabricated as approved by the Engineer.	Nos.	03		
5	Lightning Arrester	Copper rod height 2 m to be fixed on these Five Site on required height with poles complete in all respect as per drawing etc as per international standard safety or NPERA/IESCO.	Nos.	5		
		Top				
6	0.4kV LV Panel (1x1500 Kw,1x150kW & 1x100kW)	**Details of breakers and SPD to be in housed in cabinet: for AC and DC System** As given in AC & DC BoQ and drawings attached	Nos.	3		

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7	DC Distribution Box	IP65 GI DB for each inverter for placing DC strings Expected size but not limited to W*H*D = 3*1.5*0.75(16SWG) wall mount, single door with lock and with complete fitting scheme/ railing for DC Breakers	Nos.	03		
8	DC Breakers	As per Hybrid configuration and PV Module fuse ratings, as per NEPRA and International Standard or complete in all respect.	Nos.	As per inverter Hybrid Configuration		
9	DC wire for PV	6 mm Sq. DC flexible wire 1500V DC Copper conductor, Cu XLPO for array to DB(Double Insulated)	M	7000		
10	DC wire for PV	25 mm Sq. DC flexible wire 1500V DC Copper conductor, Cu XLPO from array DB to Main DB(Double Insulated)	M	250		
12	DC wire for PV	16 mm Sq. DC flexible wire 1500V DC Copper conductor, Cu XLPO from array DB to Main DB(Double Insulated)	M	250		
12	AC wire from LV panel to Inverter	Cu, XLPE/PVC 0.6/1 kV, 4*120 mm Sq. AC stranded(Double Insulated)	M	120		
13	AC wire for inverter grounding	Cu, PVC 0.6/1 kV, 1*70 mm Sq. AC flexible wire stranded	M	150		
14	DC wire for PV panel and PV frames to grounding point	PVC 1*16 mm Sq. AC standard Wire	M	600		
15	Copper Bare Conductor	25 mm Sq	M	100		
16	MC4 connectors	IP67/IP68 for PV Hybriding as per International Standard complete in all respect.	Nos.	200		

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17	Bi-directional Meter for Net Metering	Delivery of Bi-directional Meter in line with NEPRA / IESCO requirements	Nos.	03		
18	DC Earthing	Bore Type Earthing for DC Components as Per NEPRA or international Standard of safety	Nos.	03		
19	AC Earthing	Bore Type Earthing for AC Components as Per NEPRA Standard or international Standard of safety	Nos.	06		
20	Motorized Panel	• 6 x Motorized Panel Cleaners (battery operated) with minimum7-meter pole length complete in all respect.	1	Job		
Subtotal B (inclusive of all applicable taxes including GST of Services on items listed at Sr. No. 1 to 20) = PKR						
C. Installation, Testing, Commissioning, O&M Services and Training						
1	Installation, Testing and Commissioning	Complete project installation and commissioning till handover of the Project to Employer	Job/No's	03		
2	Net Billing services	Net Billing services complete in all aspects including NEPRA license and load flow studies required by the distribution company.	Job/No's	03		
3	Operation and Maintenance Services	• Cleaning of all PV Modules on each Saturday / Sunday for 06 months. • Monitoring of System Performance and submission of monthly electricity generation reports	Job/No's	02		

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4	O&M Training	1 week O & M training of 3 personnel of the Employer	Job/No's	01		
5	MISC ITEMS	• Cable ties, rawal bolts, pvc tapes, steel and PVC clamps, pvc pipes, glands different sizes, thimbles, connectors, warning labels etc. • 2 x insulated glove pairs, shoe pairs, hats for site inspection and maintenance activities • PV module O&M Toolkit 1 x 55 inch 4K TV Screen for monitoring of generation data from all inverters complete in all respect.	Job/No's	01		
Subtotal C (inclusive of all applicable taxes including GST of Services at Sr. No. 1 to 5) = PKR						
	D. Total Bid Price in PKR..... (D = A + B+C)					

Note: Value under "Subtotal B" Must NOT be less than 5% of the "Total Bid Price in PKR". Financial Bid not following the same shall be rejected.

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FORM OF BID SECURITY
(Bank Guarantee)

Security Executed on _____
(Date)

Name of Surety (Bank) with Address:
(Scheduled Bank in Pakistan)

Name of Principal (Bidder) with Address: _____

Security Amount in Rupees. _____(Rs. _____)

Bid Reference No. _____

KNOW ALL MEN BY THESE PRESENTS, that in pursuance of the terms of the Bid and at the request of the said Principal (Bidder)we, the Surety above named, are held and firmly bound unto _____
(hereinafter called the 'Employer') in the sum stated above for the payment of which sum well and truly to be made, we bind ourselves, our heirs, executors, administrators and successors, jointly and severally, firmly by these presents.

THE CONDITION OF THIS OBLIGATION IS SUCH, that whereas the Bidder has submitted the accompanying Bid dated _____ for Bid No. _____ for_____(Particulars of Bid) to the said Employer; and

WHEREAS, the Employer has required as a condition for considering said Bid that the Bidder furnishes a Bid Security in the above said sum from a Scheduled Bank in Pakistan, to the Employer, conditioned as under:

(1) that the Bid Security shall remain in force up to and including the date 30 days after the deadline for validity of bids as stated in the Instructions to Bidders or as it may be extended by the Employer, notice of which extension(s) to the Surety is hereby waived;

(2) that the Bid Security of unsuccessful Bidders will be returned by the Employer after expiry of its validity or upon signing of the Contract Agreement; and

(3) that in the event of failure of the successful Bidder to execute the proposed Contract Agreement for such work and furnish the required Performance Security, the entire said sum be paid immediately to the said Employer pursuant to Clause 15.6 of the Instruction to Bidders for the successful Bidder's failure to perform.

NOW THEREFORE, if the successful Bidder shall, within the period specified therefore, on the prescribed form presented to him for signature enter into a Contract Agreement with the said Employer in accordance with his Bid as accepted and furnish within seven(07) working days of issuance of Letter of Acceptance, a Performance Security with good and sufficient surety, as may be required, upon the form prescribed by the said Employer for the faithful performance and proper fulfillment of

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the said Contract Agreement or in the event of non-withdrawal of the said Bid within the time specified for its validity then this obligation shall be void and of no effect, but otherwise to remain in full force and effect.

PROVIDED THAT the Surety shall forthwith pay the Employer the said sum upon first written demand of the Employer (without cavil or argument) and without requiring the Employer to prove or to show grounds or reasons for such demand, notice of which shall be sent by the Employer by registered post duly addressed to the Surety at its address given above.

PROVIDED ALSO THAT the Employer shall be the sole and final judge for deciding whether the Principal (Bidder) has duly performed his obligations to sign the Contract Agreement and to furnish the requisite Performance Security within the time stated above, or has defaulted in fulfilling said requirements and the Surety shall pay without objection the said sum upon demand from the Employer forthwith and without any reference to the Principal (Bidder) or any other person.

IN WITNESS WHEREOF, the above bounden Surety has executed this Guarantee under its seal on the date indicated above, the name and seal of the Surety being hereto affixed and these presents duly signed by its undersigned representative pursuant to authority of its governing body.

SURETY (Bank)

WITNESS: Signature _____

1. _____ Name _____

_____ Title _____

Corporate Secretary (Seal) Corporate Guarantor (Seal)

2. _____

_____

Name, Title & Address

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FORM OF PERFORMANCE SECURITY
(Bank Guarantee)

Guarantee No._____
Executed on _____
Expiry date _____

[Letter by the Guarantor to the Employer]

Name of Guarantor (Bank) with address:
_____
(Scheduled Bank in Pakistan)

Name of Principal (Contractor) with address:
_____
_____

Security Amount (express in words and figures) _____
_____

Letter of Acceptance No. _____Dated _____

KNOW ALL MEN BY THESE PRESENTS, that in pursuance of the terms of the Bidding Documents and above said Letter of Acceptance (hereinafter called the Documents) and at the request of the said Principal we, the Guarantor above named, are held and firmly and irrevocably and unconditionally bound unto the _____ (hereinafter called the Employer) in the sum of the amount stated above for the payment of which sum well and truly to be made to the said Employer, we bind ourselves, our heirs, executors, administrators and successors, jointly and severally, firmly by these presents.

THE CONDITION OF THIS OBLIGATION IS SUCH, that whereas the Principal has accepted the Employer's above said Letter of Acceptance for _____ _____ (Name of Contract) for the _____

_____(Name of Project).

NOW THEREFORE, if the Principal (Contractor) shall well and truly perform and fulfil all the undertakings, covenants, terms and conditions of the said Documents during the original terms of the said Documents and any extensions thereof that may be granted by the Employer, with or without notice to the Guarantor, which notice is, hereby, waived and shall also well and truly perform and fulfill all the undertakings, covenants terms and conditions of the Contract Agreement and of any and all modifications of said Documents that may hereafter be made, notice of which modifications to the Guarantor being hereby waived, then, this obligation to be void; otherwise to remain in full force and virtue till all requirements of Clause 8.3, Defects Liability, of Conditions of Contract are fulfilled.

Our total liability under this Guarantee is limited to the sum stated above and it is a condition of any liability attaching to us under this Guarantee that the claim for payment in writing shall be received by us within the validity period of this Guarantee, failing which we shall be discharged of our liability, if any, under this Guarantee.

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We, _____ (the Guarantor), waiving all objections and defences under the Contract, do hereby irrevocably and independently guarantee to pay to the Employer without delay upon the Employer's first written demand without cavil or arguments and without requiring the Employer to prove or to show grounds or reasons for such demand any sum or sums up to the amount stated above, against the Employer's written declaration that the Principal has refused or failed to perform the obligations under the Contract Agreement which payment will be effected by the Guarantor to Employer's designated Bank & Account Number.

PROVIDED ALSO THAT the Employer shall be the sole and final judge for deciding whether the Principal (Contractor) has duly performed his obligations under the Contract Agreement or has defaulted in fulfilling said obligations and the Guarantor shall pay without objection any sum or sums up to the amount stated above upon first written demand from the Employer forthwith and without any reference to the Principal or any other person.

IN WITNESS WHEREOF, the above-bounden Guarantor has executed this Guarantee under its seal on the date indicated above, the name and corporate seal of the Guarantor being hereto affixed and these presents duly signed by its undersigned representative, pursuant to authority of its governing body.

_____
Guarantor (Bank)

Witness:

1. _____ Signature _____

_____ Name_____
Corporate Secretary (Seal)

Title_____

2. _____

_____ _____

Name, Title & Address Corporate Guarantor(Seal)

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FORM OF ADVANCE PAYMENT
(Bank Guarantee)

Guarantee No._____Date _____
WHEREAS _____(hereinafter called the 'Employer') has entered into a Contractfor _____
(Particulars of Contract) with _____ (hereinafter called the "Contractor').

AND WHEREAS, the Employer has agreed to advance to the Contractor, at the Contractor's request, an amount of Rupees _____(Rs _____) which amount shall be advanced to the Contractor as per provisions of the Contract.

AND WHEREAS, the Employer has asked the Contractor to furnish Guarantee to secure the advance for the performance of his obligations under the said Contract.

AND WHEREAS, _____
(Scheduled Bank in Pakistan) (hereinafter called the "Guarantor") at the request of the Contractor and in consideration of the Employer agreeing to make the above advance to the Contractor, has agreed to furnish the said Guarantee.

NOW, THEREFORE, the Guarantor hereby guarantees that the Contractor shall use the advance for the purpose of above-mentioned Contract and if he fails and commits default in fulfilment of any of his obligations for which the advance payment is made, the Guarantor shall be liable to the Employer for payment not exceeding the aforementioned amount. Notice in writing of any default, of which the Employer shall be the sole and final judge, on the part of the Contractor, shall be given by the Employer to the Guarantor, and on such first written demand, payment shall be made by the Guarantor of all sums then due under this Guarantee without any reference to the Contractor and without any objection. This Guarantee shall remain in force until the Delivery of Machinery, Equipment and Material of the System at Site upon verification and confirmation of the Employer. The Guarantor's liability under this Guarantee shall not in any case exceed the sum of Rupees _____(Rs _____). This Guarantee shall remain valid up to seven (07) days of the aforesaid confirmation letter by the Employer and shall be null and void after the aforesaid date.

GUARANTOR

Signature _____
Name / Title _____

WITNESS
1. _____
_____
Corporate Secretary (Seal)
2. _____
(Name Title & Address)
Corporate Guarantor(Seal)

_____

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FORM OF CONTRACT AGREEMENT

THIS CONTRACT AGREEMENT (hereinafter called the " Contract Agreement") made on the _____ day of_____(month) 20_____ between _____ (hereafter called the "Employer") of the one part and _____ (hereafter called the "Contractor") of the other part.

WHEREAS the Employer is desirous that certain Works, viz _____ should be executed by the Contractor and has accepted a Bid by the Contractor for the execution and completion of such Works and the remedying of any defects therein.

NOW this Agreement witnessed as follows:

1. In this Agreement words and expressions shall have the same meanings as are respectively assigned to them in the Conditions of Contract hereinafter referred to.

2. The following documents after incorporating addenda, if any, except those parts relating to Instructions to Bidders shall be deemed to form and be read and construed as part of this Contract Agreement, viz:

   (a) The Contract Agreement;
   (b) Letter of Acceptance;
   (c) Specifications and Employer's Design
   (d) Contract Data;
   (e) Conditions of Contract;
   (f) Letters of Technical Bid and Price Bid;
   (g) The priced Bill of Quantities; and
   (h) The Appendices to Bid.

3. In the event of any discrepancy or inconsistency within the Contract Documents, then the documents shall prevail in the order listed above.

4. In consideration of the payments to be made by the Employer to the Contractor as hereinafter mentioned, the Contractor hereby covenants with the Employer to execute and complete the Works and remedy defects therein in conformity and in all respects with the provisions of this Contract Agreement.

5. The Employer hereby covenants to pay the Contractor, in consideration of the execution and completion of the Works as per provisions of this Contract Agreement, the Contract Price or such other sum as may become payable under the provisions of this Contract Agreement at the times and in the manner prescribed therein.

IN WITNESS WHEREOF the parties hereto have caused this Contract Agreement to be executed on the day, month and year first before written in accordance with their

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respective laws.

Signature of the Contactor	Signature of Employer
_____	_____
(Seal)	(Seal)
In the presence of:	
Witness:	Witness:
_____	_____
(Name, Title and Address)	(Name, Title and Address)

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Letter of Acceptance

[letterhead paper of the Employer]

NAME OF CONTRACT:_____

CONTRACT NUMBER:_____

TO :_____

Date:_____

Your Reference:_____

Our Reference:_____

We thank you for your Bid dated _____ for the execution and completion of the Works comprising the above-named Contract and remedying of defects therein, all in conformity with the terms and conditions contained in the Contract.

We have pleasure in accepting your Bid for the Accepted Contract Amount of:

_____
[currency and amount in figures]

_____
[currency and amount in words]

In consideration of you properly and truly performing the Contract, we agree to pay you the Accepted Contract Amount or such other sums to which you may become entitled under the terms of the Contract, at such times and as prescribed by the Contract.

We acknowledge that this Letter of Acceptance creates a binding Contract between us, and we undertake to fulfil all our obligations and duties in accordance with the terms of this Contract.

Signature: _____

Signed by: _____

For and on behalf of:_____

Date:_____

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SPECIFICATIONS

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SPECIFICATIONS

The following are the minimum technical specifications of the equipment being parts of the Solar PV System that shall be installed at the premises.

1. PV Modules

Solar PV modules shall be of Tier-1 manufacturer as per latest Bloomberg New Energy Finance (BNEF) list (for Q1 or Q2 of the year 2026). The PV technology shall be half-cut N-type or better to maximize the yield against the available area and minimize the losses against partial shading. Each module shall have atleast IP 68 rated Junction Boxes which shall be dust and vermin proof having copper bus bar terminals, EPDM rubber gasket, MOVs and reverse blocking Diodes and IEC 62790 compliant.

a) Performance and Rating Requirements:

The power rating for each PV module shall be more than 650Wp under STC conditions with positive power tolerance only. Each PV module must have efficiency of equal to or greater than 23%. The proposed PV module must exhibit efficiency of more than 96% of the STC efficiency at 200 W/m^2 (i.e. low lighting conditions). The module Temperature Coefficient at Pmax shall be 0.3% / ^0C or better in performance.

b) Certification Requirements:

The PV Module must be Type Tested based on IEC standard 61215 (Terrestrial PV Modules- Design qualification and type approval) and IEC 61730 (Photovoltaic module safety qualification) based insulation of Safety Class II. Modules must be free of Potential Induced Degradation (PID) and Light Induced Degradation (LID). The module shall be S.R.O 604 (or equivalent) compliant issued by Government of Pakistan. The Flash Test report, module Test report as per IEC standards 61215 and 61730 along with pre-shipment inspection certificate shall be provided at the time of delivery of modules at site. Quality assurance/Quality control certification for adherence as per European standards.

c) Warranty and Replacement:

Each module(panel) shall have minimum 12 years' material and replacement warranty and 25 years' performance warranty. The Power output of each PV module shall be greater than 85% after 25 Years.

d) Product Identification:

Each PV module shall have a unique barcode identification tag issued by the manufacturer. The same shall be confirmed from the manufacturer prior to verification of equipment delivery at site by the third party.

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2. Inverters:

a) Performance and Rating Requirements:

The inverter shall be three phase grid tied with IP66 protection having CEC efficiency of not less than 98.5 % and output frequency of 50 to 60 Hz. The cumulative capacity of inverter shall be according to the total solar PV capacity to be installed and inverter's minimum input voltage shall be as per array specific design. Under all scenarios, the DC to AC ratio of the inverter shall not be more than 1.1.

The inverters shall have multiple MPPT inputs for 3-phase inverters. The Total Harmonic Distortion shall be less than 3%. The ambient operating temperature range of Inverter shall be between -25°C to 60°C. Three phase inverter must have Diesel Generator Synchronization module for seamless integration with Generators.

(b) Protection and Safety Requirements:

The inverter shall have AC overcurrent protection, residual current monitoring, DC reverse polarity protection, DC insulation resistance detection, PV array Hybrid fault monitoring, anti- islanding protection and Type-2 or better DC and AC SPDs. The inverter must have anti PID module to limit the harmful effects of Potential Induced Degradation of PV Modules. Inverter shall have a built-in data logger; communication interface protections and remote monitoring capability. It shall be compliant with UL 1741, IEC 62109-1/2 for safety design.

Warranty Requirements:

The inverters shall have minimum 12 years of replacement warranty.

3. PV Balance of system:

a) Mounting Structure:

The mounting structure shall be UV resistant and shall have hot dip galvanized steel frame of gauge 12 with 120 micron or thicker zinc coating. The structure shall have galvanic isolation and shall withstand wind speed of 160 km/h. complete civil work (including PCC (1:2:4) of 3,000 psi strength) for mounting structure installation shall be performed as per drawing attached.

Design requirements for mounting structure:

The supporting structure including PV modules shall be capable of handling the maximum 3 sec- gust wind speed of up to 160 km/h. The EPC contractor shall provide a system design that meets this requirement as well as any additional requirement related to the site-specific snow load (if applicable). Ballasting/concrete block sizing calculations are typically made by the use of professional design software and shall be made available for the installation and plant documentation.

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All components of the mounting system and all connecting materials shall be made of corrosion- proof material. All fastening structures shall be verified for durability and resistance to environmental degradation for 25 years, the lifetime requirement of the complete structure. The structural elements shall consist of anodized aluminum, galvanized steel as per ASTM A123/A 123M, or materials with equivalent if not better properties. The applicable building code / construction codes shall be used as well.

The mounting structures shall be earthed for maximum short-circuit current and lightning protection. A racking design that removes the need of PV module to PV module grounding by using the common rail / PV module clips for the grounding is recommended.

Dismantling of existing sheds and site preparatory works and back fillings shall be performed to the satisfaction of Client.

a) DC Cables, AC Cables, Earthing wires, Conduits, Connectors, Cable Trays etc.

Copper cables with purity of 99.9% or more with XLPO double insulation and rated for 1,500 V shall be used. Cables shall be low smoke Halogen free along with compliance with BS 6004, BS 7644 and UL 4703 standards. Minimum size for DC cables shall be 6 mm^2 or as per size of PV Module Output cable (whichever is higher or given in the Bills of Quantities). Under all conditions, for DC cables, voltage drop shall be less than 0.5% at STC and for AC cables voltage drop shall be less than 1% at STC. 6 mm^2 earth wires for equipotential bonding shall be used. Separate DC and AC earthing each with 3 ohm or lesser resistance shall be achieved using the earthing arrangement.

Perforated cable trays of 16 gauge or thicker and UPVC/corrugated PVC pipes shall be installed with seamless terminations. The cable tray shall not be physically accessible for routine works. Flexible pipes, glands, conduits, cables ducts shall be installed as per Employer Design. Conduits shall have 30% or higher spare capacity. Labeling of all cables shall be performed in compliance with IEC Standard 62491.

b) Breakers, Surge Arrester, LV Panel etc.

Circuit breaker, Protection devices and Non-Isolated Lightning Arrestors shall be installed as per Employer Design. MCB/MCCB Circuit breaker for each Hybrid shall be rated for more than 1.5 times the STC current rating of the Hybrid. Under all circumstances circuit breaker rating must be less than the fuse rating of PV Module.

In case of installation of LV panel, the panel shall be IP54 rated of galvanized material with at least 16 gauge. The panel shall have protection scheme compliant with IEC Standard 62548, input and output circuit breakers as per IEC Standards, bus bar current bearing capacity as per requirement, and isolators for bus bars and AC grounding bus bar. The panel shall have digital Ammeter and Voltmeter, RYB indicator with breather vents on both sides of cabinets with RAL Color: 6021.

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c) Monitoring and Control System:

Online and real time monitoring shall be provided with following parameters;

1. Energy generation (kWh)
2. Power production (kW)
3. Hybrid-level monitoring of Voltage and Current
4. Storage and monitoring data for at least 3 years
5. TV Screen of size mentioned in BoQ

d) Net Metering Application and Installation of Bidirectional Meter

The Contractor shall comply with all prevailing regulations set forth by NEPRA for net metering purposes including load flow studies on requisite software. Preparation and processing of application for net metering along with requisite clarifications shall be arranged by the Contractor in the Project timelines. Furthermore, in cases where the technical feasibility study is required for purposes of approval of net-metering application, the Contractor shall also carry out the requisite feasibility study and make arrangement for its approval from the relevant DISCO.

e) Operation Manual:

An Operation, Instruction and Maintenance Manual, in English shall be provided with the Solar PV System. The detailed diagram(s) of wiring and connection diagrams shall also be provided with the manual.

Note:

- Various components of Solar PV System shall additionally be conformed to the relevant national/international Electrical Safety Standards wherever applicable.

4. Operation and Maintenance

Following the successful Taking Over of the Project, the EPC Contractor shall provide PV Module Cleaning services on each Saturday/Sunday and submit monthly electricity generation reports of PV System for duration of 06 months.

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COMSATS UNIVERSITY ISLAMABAD (CUI) ATTOCK CAMPUS DESIGN DRAWINGS

The CUI Design Drawings of the System is attached hereto.

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PUBLIC PROCUREMENT REGULATORY AUTHORITY
(PPRA)
EPADS

CONSTRUCTION EXPERIENCE

Form EXP - 4.1
General Construction Experience

Bidder's Name: _____
Date: _____
JV Member's Name_____
Bid Reference No. (if any) and title: _____
Page _____of _____pages

Starting Year	Ending Year	Contract Identification	Role of Bidder
		Contract name: _____ Brief Description of the Works performed by the Bidder: _____ Amount of contract: _____ Name of EMPLOYER: _____ Address: _____	
		Contract name: _____ Brief Description of the Works performed by the Bidder: _____ Amount of contract: _____ Name of EMPLOYER: _____ Address: _____	
		Contract name: _____ Brief Description of the Works performed by the Bidder: _____ Amount of contract: _____ Name of EMPLOYER: _____ Address: _____	

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Form EXP - 4.2(a)

Specific Construction and Contract Management Experience

Bidder's Name: _____
Date: _____
JV Member's Name_____
Bid Reference No. (if any) and title: _____
Page _____of _____pages

Similar Contract No.	**Information**			
Contract Identification				
Award date				
Completion date				
Role in Contract	Prime Contractor ☐	Member in JV ☐	Management Contractor ☐	Sub-contractor ☐
Total Contract Amount			PKR equivalent	
If member in a JV or sub-contractor, specify participation in total Contract amount				
EMPLOYER's Name:				
Address: Telephone/fax number E-mail:				

Form EXP - 4.2(a) (cont.)
Specific Construction and Contract Management Experience (cont.)

Similar Contract No.	**Information**
Description of the similarity in accordance with Sub-Factor 4.2(a) of Section III:	
1. Amount	
2. Physical size of required works items	
3. Complexity	
4. Methods/Technology	
5. Construction rate for key activities	
6. Other Characteristics	

PUBLIC PROCUREMENT REGULATORY AUTHORITY (PPRA) EPADS

Form EXP - 4.2(b)
Construction Experience in Key Activities

Bidder's Name: _____
Date: _____
Bidder's JV Member Name: _____
Sub-contractor's Name (if any): _____
Bid Reference No. (if any) and title: _____

Page _____of _____pages

All Sub-contractors for key activities must complete the information and Qualification Criteria and Requirements, Sub-Factor 4.2.

1. Key Activity No One: _____

	Information			
Contract Identification				
Award date				
Completion date				
Role in Contract	Prime Contractor ☐	Member in JV ☐	Management Contractor ☐	Sub-contractor ☐
Total Contract Amount			PKR equivalent	
Quantity (Volume, number or rate of production, as applicable) performed under the contract per year or part of the year	Total quantity in the contract (i)	Percentage participation (ii)		Actual Quantity Performed (i) x (ii)
Year 1				
Year 2				
Year 3				
Year 4				
EMPLOYER's Name:				
Address: Telephone/fax number				

	Information
E-mail:	

2. Activity No. Two

3. ……

	Information
Description of the key activities in accordance with Section III:	

PUBLIC PROCUREMENT REGULATORY AUTHORITY (PPRA) EPADS

Form EXP - 4.2 (c)
Specific Experience in Managing ES aspects

[The following table shall be filled in for contracts performed by the Bidder, and each member of a Joint Venture]

Bidder's Name: *[insert full name]*
Date: *[insert day, month, year]*
Joint Venture Member Name: *[insertfull name]*
Bid Reference No. (if any) and title: *[insert ICB/NCB number and title]*
Page *[insert page number]* of *[insert total number]* pages

1. Key Requirement no 1 in accordance with 4.2 (c): _____

Contract Identification				
Award date				
Completion date				
Role in Contract	Prime Contractor ☐	Member in JV ☐	Management Contractor ☐	Subcontractor ☐
Total Contract Amount			PKR	
Details of relevant experience				

2. Key Requirement no 2 in accordance with 4.2 (c): _____

3. Key Requirement no 3 in accordance with 4.2 (c): _____

4. [PA define key requirements]

Contractor's Representative and Key Personnel Schedule

Bidders should provide the names and details of the suitably qualified Contractor's Representative and Key Personnel to perform the Contract. The data on their experience should be supplied using the Form PER-2 below for each candidate.

Contractor' Representative and Key Personnel

1.	**Title of position:**	
	Name of candidate:	
	Duration of appointment:	[*insert the whole period (start and end dates) for which this position will be engaged*]
	Time commitment: for this position:	[*insert the number of days/week/months/ that has been scheduled for this position*]
	Expected time schedule for this position:	[*insert the expected time schedule for this position (e.g. attach high level Gantt chart*]
2.	**Title of position:**	
	Name of candidate:	
	Duration of appointment:	[*insert the whole period (start and end dates) for which this position will be engaged*]
	Time commitment: for this position:	[*insert the number of days/week/months/ that has been scheduled for this position*]
	Expected time schedule for this position:	[*insert the expected time schedule for this position (e.g. attach high level Gantt chart*]
3.	**Title of position:**	
	Name of candidate:	
	Duration of appointment:	[*insert the whole period (start and end dates) for which this position will be engaged*]
	Time commitment: for this position:	[*insert the number of days/week/months/ that has been scheduled for this position*]
	Expected time schedule for this position:	[*insert the expected time schedule for this position (e.g. attach high level Gantt chart*]
4.	**Title of position:**	
	Name of candidate:	
	Duration of appointment:	[*insert the whole period (start and end dates) for which this position will be engaged*]

	Time commitment: for this position:	[*insert the number of days/week/months/ that has been scheduled for this position*]
	Expected time schedule for this position:	[*insert the expected time schedule for this position (e.g. attach high level Gantt chart*]
5.	**Title of position:**	
	Name of candidate	
	Duration of appointment:	[*insert the whole period (start and end dates) for which this position will be engaged*]
	Time commitment: for this position:	[*insert the number of days/week/months/ that has been scheduled for this position*]
	Expected time schedule for this position:	[*insert the expected time schedule for this position (e.g. attach high level Gantt chart*]
6.	**Title of position:** *[insert title]*	
	Name of candidate	
	Duration of appointment:	[*insert the whole period (start and end dates) for which this position will be engaged*]
	Time commitment: for this position:	[*insert the number of days/week/months/ that has been scheduled for this position*]
	Expected time schedule for this position:	[*insert the expected time schedule for this position (e.g. attach high level Gantt chart*]

Form PER-2:
Resume and Declaration
Contractor's Representative and Key Personnel

Name of Bidder

Position [*#1*]: [*title of position from Form PER-1*]		
Personnel information	**Name:**	**Date of birth:**
	Address:	**E-mail:**
	Professional qualifications:	
	Academic qualifications:	
	Language proficiency:*[language and levels of speaking, reading and writing skills]*	
details		
	Address of Employer:	
	Telephone:	**Contact (manager / personnel officer):**
	Fax:	
	Job title:	**Years with present Employer:**

Summarize professional experience in reverse chronological order. Indicate particular technical and managerial experience relevant to the project.

Project	**Role**	**Duration of involvement**	**Relevant experience**
[main project details]	*[role and responsibilities on the project]*	*[time in role]*	*[describe the experience relevant to this position]*

Declaration

I, the undersigned *[insert either "Contractor's Representative" or "Key Personnel" as applicable]* , certify that to the best of my knowledge and belief, the information contained in this Form PER-2 correctly describes myself, my qualifications and my experience.

I confirm that I am available as certified in the following table and throughout the expected time schedule for this position as provided in the Bid:

Commitment	**Details**
Commitment to duration of contract:	*[insert period (start and end dates) for which this Contractor's Representative orKey Personnel is available to work on this contract]*
Time commitment:	*[insert period (start and end dates) for which this Contractor's Representative orKey Personnel is available to work on this contract]*

I understand that any misrepresentation or omission in this Form may:

(a) be taken into consideration during Bid evaluation;

(b) result in my disqualification from participating in the Bid;

(c) result in my dismissal from the contract.

Name of Contractor's Representative or **Key Personnel: [*insert name*]**

Signature: _____

Date: (day month year): _____

Countersignature of authorized representative of the Bidder:

Signature: _____

Date: (day/month/year): _____

Financial Resources

Specify proposed sources of financing, such as liquid assets, unencumbered real assets, lines of credit, and other financial means, net of current commitments, available to meet the total construction cash flow demands of the subject contract or contracts as specified in Eligibility and Qualification Criteria.

No.	**Source of financing**	**Amount (Eq. PKR)**
1		
2		
3		

Past Experience / Contracts

Contracts over *[insert amount]* during the last three years:				
Procuring Agency	Value	Year	Goods/Services Supplied	Country of Destination

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Historical Contract Non-Performance, and Pending Litigation and Litigation History

[The following table shall be filled in for the Applicant and for each member of a Joint Venture]

Applicant's Name: *[insert full name]*
Date: *[insert day, month, year]*
Joint Venture Member Name: *[insert full name]*
IFP No. and title: *[insert IFP number and title]*
Page *[insert page number]* of *[insert total number]* pages

☐ Not debarred due to deviation from commitment of Bid Securing Declaration-☐ Not debarred due to non-performance			
Year	**Non-performed portion of contract**	**Contract Identification**	**Total Contract Amount (current value, currency, exchange rate and PKR equivalent)**
[insert year]	*[insert amount and percentage]*	Contract Identification: *[indicate complete contract name/ number, and any other identification]* Name of Procuring Agency: *[insert full name]* Address of Procuring Agency: *[insert street/city/country]* Reason(s) for nonperformance: *[indicate main reason(s)]*	*[insert amount]*
Pending Litigation, in accordance with Section III, Qualification Criteria and Requirements			
☐ Pending litigation in accordance with Section III, Qualification Criteria and Requirements, Sub-Factor 2.3 as indicated below.			

Year of dispute	**Amount in dispute (currency)**	**Contract Identification**	**Total Contract Amount (currency), ~~US$~~ PKR Equivalent (exchange rate)**

[insert year]	*[insert amount]*	Contract Identification: [indicate complete contract name, number, and any other identification] Name of Procuring Agency: *[insert full name]* Address of Procuring Agency: *[insert street/city/country]* Matter in dispute: *[indicate main issues in dispute]* Party who initiated the dispute: *[indicate "Procuring Agency" or "Supplier"]* Status of dispute: *[Indicate if it is being treated by the Adjudicator, under Arbitration or being dealt with by the Judiciary]*	*[insert amount]*

☐ No consistent history of court/arbitral award decisions in accordance with Section III, Qualification Criteria and Requirements, Sub-Factor 2.4.

☐ Consistent history of court/arbitral award decisions in accordance with Section III, Qualification Criteria and Requirements, Sub-Factor 2.4 as indicated below.

Year of award	**Outcome as percentage of Net Worth**	**Contract Identification**	**Total Contract Amount (currency), PKR Equivalent (exchange rate)**
[insert year]	*[insert percentage]*	Contract Identification: [indicate complete contract name, number, and any other identification] Name of Procuring Agency: *[insert full name]* Address of Procuring Agency: *[insert street/city/country]* Matter in dispute: *[indicate main issues in dispute]* Party who initiated the dispute: *[indicate "Procuring Agency" or "Supplier"]* Court/ arbitral award decision: *[Indicate if the award decision was against the Applicant or any member of a joint venture.]y]*	*[insert amount]*

Current Contract Commitments / Works in Progress

Bidders and each member to a JV should provide information on their current commitments on all contracts that have been awarded, or for which a letter of intent or acceptance has been received, or for contracts approaching completion, but for which an unqualified, full completion certificate has yet to be issued.

Current Contract Commitments					
No.	Name of Contract	**Employer's Contact Address, Tel, Fax**	**Value of Outstanding Work [Current Eq. PKR]**	**Estimated Completion Date**	**Average Monthly Invoicing Over Last Six Months [Eq. PKR/month)]**
1					
2					
3					
4					
5					

Financial Situation and Performance

[The following table shall be filled in for the Applicant and for each member of a Joint Venture]

Applicant's Name: *[insert full name]*
Date: *[insert day, month, year]*
Joint Venture Member Name: *[insert full name]*
IFP No. and title: *[insert IFP number and title]*
Page *[insert page number]* of *[insert total number]* pages

1. Financial data

Type of Financial information in (currency)	**Historic information for previous _[insert number] years, [insert in words] (amount in currency, currency, exchange rate*, PKR equivalent)**				
	Year 1	Year 2	Year 3		
Statement of Financial Position (Information from Balance Sheet)					
Total Assets (TA)					
Total Liabilities (TL)					
Total Equity/Net Worth (NW)					
Current Assets (CA)					
Current Liabilities (CL)					
Working Capital (WC)					
Information from Income Statement					
Total Revenue (TR)					
Profits Before Taxes (PBT)					
Cash Flow Information					
Cash Flow from Operating Activities					

* Refer ITA 14 for the exchange rate

3. Financial documents

The Applicant and in case of JV, members of JV shall provide copies of financial statements for *[number]* years pursuant Section III, Qualifications Criteria and Requirements. The financial statements shall:

(a) reflect the financial situation of the Applicant or in case of JV member, and not an affiliated entity (such as parent company or group member).

(b) be independently audited or certified in accordance with local legislation.

(c) be complete, including all notes to the financial statements.

(d) correspond to accounting periods already completed and audited.

☐ Attached are copies of financial statements[1] for the *[number]* years required above; and complying with the requirements.

[1] If the most recent set of financial statements is for a period earlier than 12 months from the date of Application, the reason for this should be justified.

Average Annual Turnover (Annual Sales Value)

[The following table shall be filled in for the Applicant and for each member of a Joint Venture]

Applicant's Name: *[insert full name]*
Date: *[insert day, month, year]*
Joint Venture Member Name: *[insert full name]*
IFP No. and title: *[insert IFP number and title]*
Page *[insert page number]* of *[insert total number]* pages

Annual Turnover Data			
Year	**Amount** **Currency**	**Exchange rate*** **(If applicable)**	**PKR equivalent**
[indicate calendar year]	*[insert amount and indicate currency]*		
		Average Annual Turnover **	

* Refer ITA for date and source of exchange rate.

** Total PKR equivalent for all years divided by the total number of years. See Section III, Qualification Criteria and Requirements, ITA.